LAB 7 REPORT

Fuzzing for Input Validation Bugs

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Section	D
Course/Lab	Software Engineering Lab 7

Objective

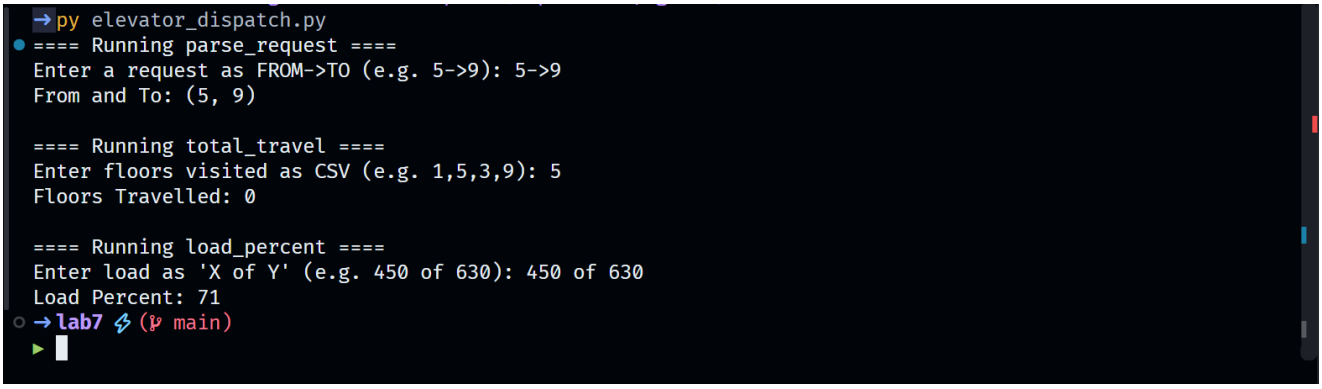
Use Hypothesis-based fuzz testing to generate unexpected inputs, identify input-validation failures, and improve the robustness of the elevator dispatch application.

Application functions tested

parse_request(), total_travel(), and load_percent(). The test suite includes generated string and None inputs, valid input properties, and selected invalid-input checks.

1. Application Execution

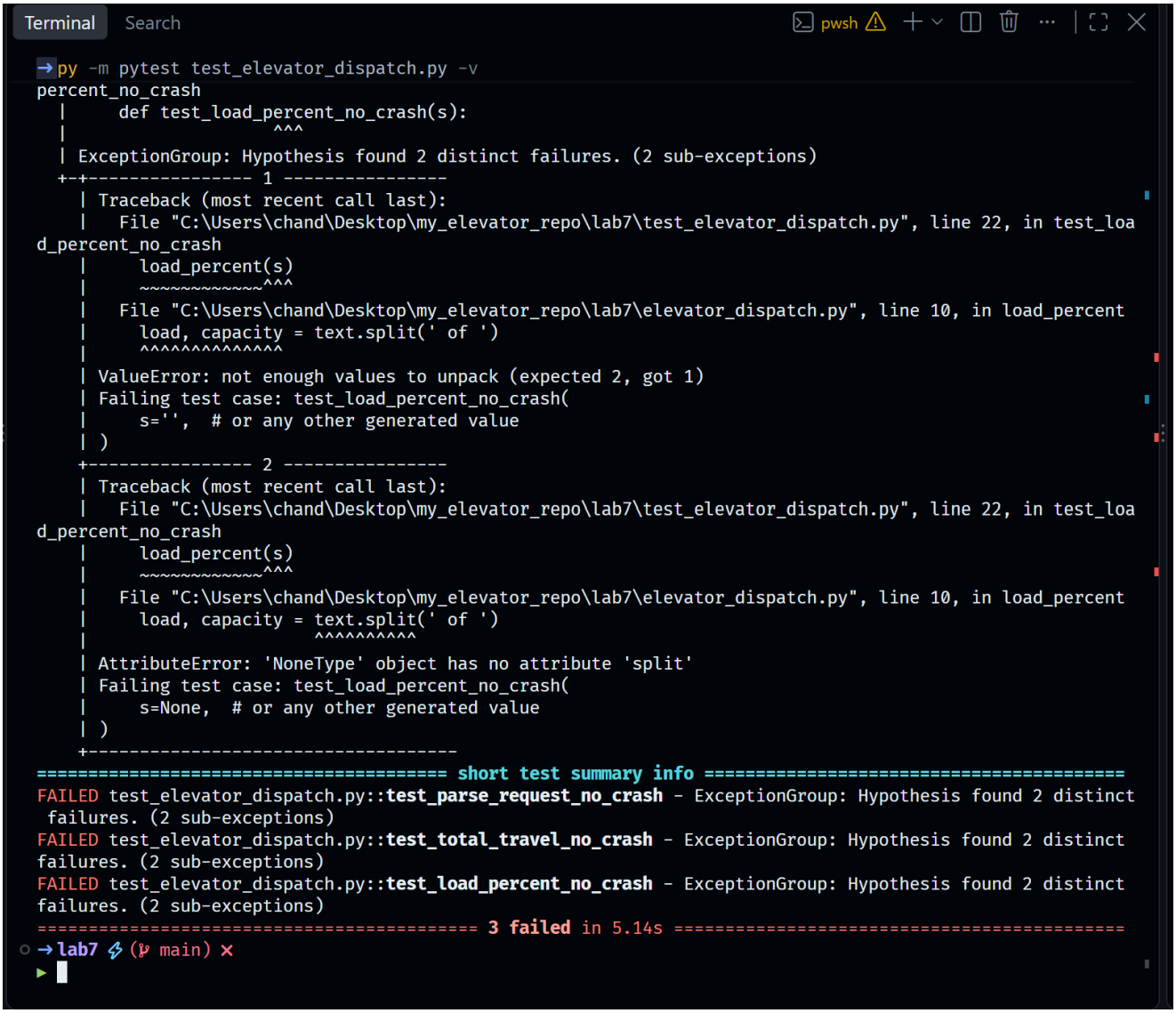
The application was run interactively with sample elevator request and load inputs. In this captured run, the travel input was a single floor value (5), so the displayed travel distance was 0. The later corrected run uses the multi-floor sequence 1,5,3,9.



Screenshot 1 of 4

2. Fuzz Testing: Failures Identified

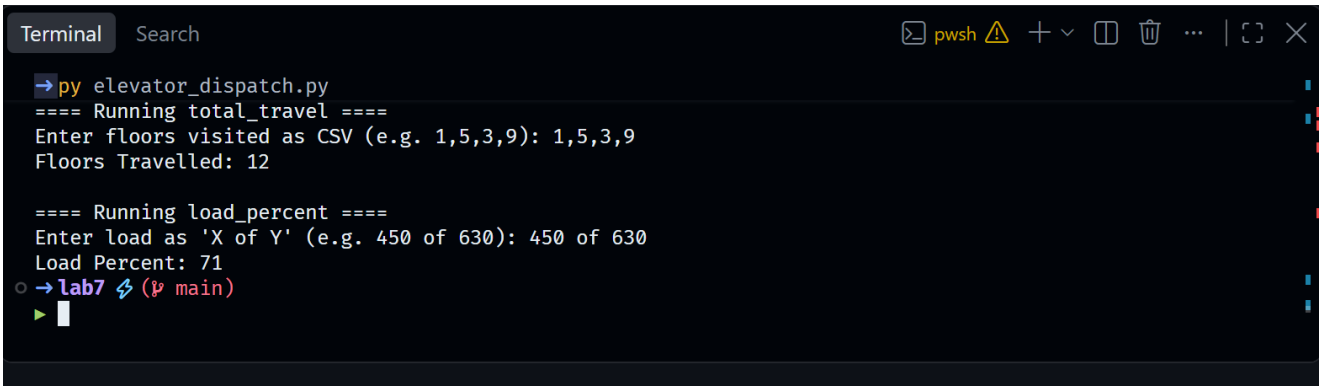
The initial Hypothesis tests exposed unhandled inputs, including an empty string and None. The captured traceback shows failures in load_percent() when the implementation attempted to split invalid input. These failures motivated input validation in the application.



Screenshot 2 of 4

3. Fixed Application Output

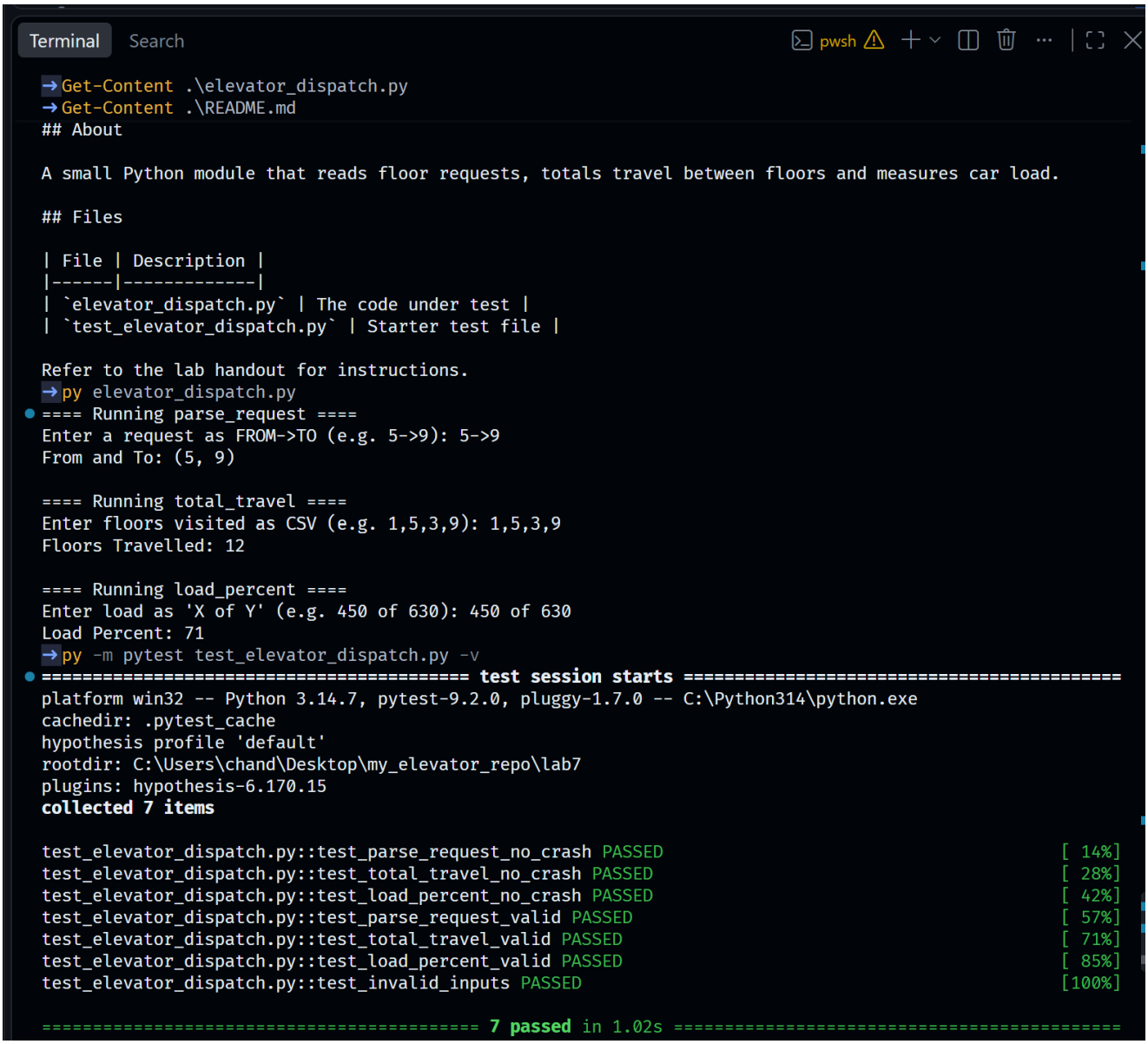
After updating the application, the sample request 5->9 was parsed successfully, the floor sequence 1,5,3,9 produced a travel distance of 12, and 450 of 630 produced a load percentage of 71.



Screenshot 3 of 4

4. Fuzz Testing: Successful Results

The updated Hypothesis test suite completed successfully. The terminal output reports seven tests passed, covering generated inputs, valid requests, travel calculations, load percentages, and selected invalid inputs.



Screenshot 4 of 4